Analytic parameter sensitivities in ManifoldsX: scope and validation

manifoldsx.github.io/differentiable-dynamics

ManifoldsX computes how motion and joint loads change with spring and bushing stiffness and damping, analytically, at every step of a simulated trajectory. The derivatives cover consistent initialization, the implicit index-3 HHT step and the rotation update. They were compared with independent perturbed simulations over 30-second runs. These are numerical checks of the derivatives; they say nothing about hardware accuracy.

1. What they are for

Understanding compliance

How a stiffness or damping coefficient changes displacement, velocity, acceleration and reaction forces at every instant.

Calibration and design

Gradients are the building blocks for parameter fitting and design studies. A fitting or optimization workflow for a given application still has to be built and validated.

2. Validation

Table 1. Validation results

Study	Evidence
Spring stiffness and damping	Analytic trajectory derivatives compared with Richardson-extrapolated finite differences over 30-second runs; every closed-form term also checked against finite differences.
6D bushings	All 12 diagonal coefficients; 24 trajectories of 30 seconds on grounded and moving-parent connections.
Bushing motion derivatives	Worst sampled error relative to the peak reference: 1.11×10^{-7} for position and rotation, 2.37×10^{-7} for velocity.
Bushing accelerations and loads	Same metric: 3.80×10^{-5} for acceleration, 5.47×10^{-5} for multipliers and 5.46×10^{-5} for generalized reactions. These are limited by the finite-difference reference, not by the derivatives.
Sparse sensitivity solve	Damped chains from 10 to 500 links, with one symbolic pattern analysis per run.

Errors are normalized by the peak reference value over the sampled times, not per component at every instant. The analytic runs used no finite-difference evaluations; independent perturbed simulations supplied the reference. In the bushing tests, computing sensitivities left the nominal trajectory bit-for-bit unchanged.

3. Scope

Covered

Forward sensitivities for spring stiffness and damping and the 12 diagonal bushing coefficients, one scalar parameter per sensitivity instance; bushing support is opt-in. The trajectory studies use index-3 HHT with full-rank revolute, spherical and translational joints and a parameter-independent initial pose and velocity.

Not covered yet

Contact, controllers and runtime loads, redundant or rank-changing constraints, projection and other integrators. Adaptive step decisions are not differentiated. The Stewart motion benchmark does not establish differentiability of that whole model.
